Gene Therapy: An Overview

The development of any gene therapy approach requires an accurate evaluation of a series of parameters, including assessment of the likelihood that a **given disease might realistically be treated by gene therapy**, followed by the **choice of the correct therapeutic gene**, regulatory signals, should this gene be expressed in the cells, **delivery system**, and **administration route.**

1. Choice of Therapeutic Gene

A vast spectrum of nucleic acids with potential therapeutic function is now available to gene therapy. This includes **protein-coding genes** (or, more often, cDNAs) and **a broad series of small, non-coding nucleic acids**, among which oligonucleotides, usually used as antisense to a target mRNA, ribozymes, siRNAs, RNA and DNA decoys, and aptamers.

2. Administration Route

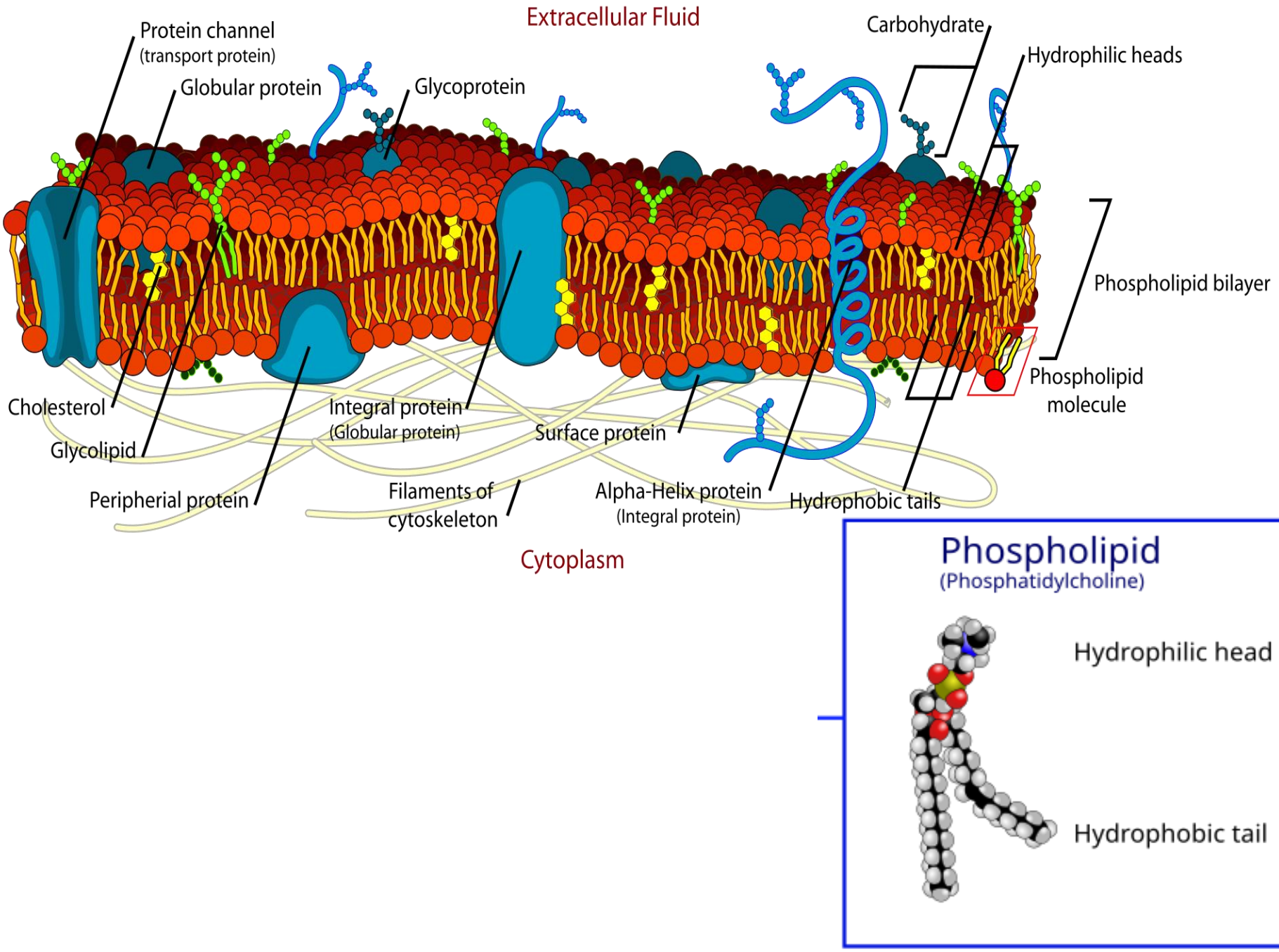
Gene therapy can follow one of two general routes of administration, based on either the isolation of the patient's cells followed by gene transfer in the laboratory (***ex vivo* gene therapy**) or on the direct delivery of the therapeutic gene into the patients (***in vivo* gene therapy**).

3. Gene Delivery System

Probably the most crucial factor conditioning the success of any gene therapy approach is the efficiency with which therapeutic nucleic acids enter the target cells. **The hydrophobic plasma membrane** of mammalian cells represents a **formidable barrier** for large polyanions such as DNA or RNA. With very few exceptions, naked nucleic acids are, therefore, very poorly internalized by the cells. Gene transfer must be facilitated using **physical** (for example, electroporation or high pressure injection), **chemical** (cationic lipids or polymers), or **biological** (viral vectors) tools.

- The cell membrane (also known as the plasma membrane or cytoplasmic membrane, and historically referred to as the plasmalemma) is a biological membrane that separates and protects the interior of a cell from the outside environment (the extracellular space) The cell membrane consists of a lipid bilayer, made up of two layers of phospholipids with cholesterols (a lipid component) interspersed between them, maintaining appropriate membrane fluidity at various temperatures..

- The membrane also contains membrane proteins, including integral proteins that span the membrane and serve as membrane transporters, and peripheral proteins that loosely attach to the outer (peripheral) side of the cell membrane, acting as enzymes to facilitate interaction with the cell's environment.[3] Glycolipids embedded in the outer lipid layer serve a similar purpose


Extracellular Fluid
Protein channel
(transport protein)
Globular protein
Glycoprotein
Carbohydrate
Hydrophilic heads
Phospholipid bilayer
Phospholipid
molecule
Cholesterol
Glycolipid
Peripherial protein
Integral protein
(Globular protein)
Filaments of
cytoskeleton
Surface protein
Alpha-Helix protein
(Integral protein)
Hydrophobic tails
Cytoplasm
Phospholipid
(Phosphatidylcholine)
Hydrophilic head
Hydrophobic tail

In the case of naked DNA, gene transfer into mammalian cells is known as **transfection**; when a viral vector is used instead, delivery of a gene is called **transduction**. At least four families of viruses are currently considered in clinical experimentations, those based on retroviruses (gammaretroviruses and lentiviruses), adenoviruses, adeno-associated viruses (AAVs), and herpesviruses. The main methods for gene delivery are described in the chapter on 'Methods for Gene Delivery'.

4. Cell Targeting

In in vivo gene therapy, it would be important that the therapeutic gene might be internalized exclusively by the desired cell type. As a matter of fact, this requirement recapitulates the notion originally put forward by the scientist and physician Paul Ehrlich, winner of the Nobel prize for Physiology and Medicine in 1908, who introduced the concept of the magic bullet, namely an ideal drug that, upon systemic injection, could selectively target a specific cellular target (which, in the concept of Ehrlich, was mainly a pathogenic microorganism).

5. Persistence of the Therapeutic Gene

Another highly important parameter to consider in gene therapy is the persistence of gene transfer. This is essentially linked to the characteristics of the nucleic acid and the delivery system used. When the therapeutic gene is a small regulatory RNA or RNA (antisense oligonucleotide, aptamers, ribozymes, siRNAs) injected systemically, disappearance from circulation and elimination via the liver, kidney, and reticulo-endothelial system is usually very rapid.

6. Expression of the Therapeutic Gene

When the therapeutic gene needs to be expressed in the target cells, as is the case for genes coding for proteins or shRNAs, an essential issue is to obtain high, persistent, and, possibly, tissue-specific levels of expression. In some situations, it is instead mandatory that expression of the therapeutic gene be more physiological.

An example of this requisite is gene therapy of thalassemias, in which synthesis of the globin chains needs to occur in the erythroblast in a balanced manner, to avoid precipitation of the proteins in excess inside the cells, determining their damage and death. Since the sequences of promoters driving expression of endogenous, regulated genes are often poorly known or, in any case, too large to be used, it is essential to identify new tissue specific promoters that are sufficiently short to be accommodated into viral vectors.